

**SAT Math Mastery · Week TP1F****SAT Trick Pack 1 — Part F****Time** 120 minutes: 90 min solving + 30 min self-check and error log**Topics** margin of error · chained percents · percent multipliers · coordinates and angles · exponential vs. linear · cubes in fractions · volume and unit cubes · Factor Theorem · recurrences · multi-ratio · bounded exponentials · altitude to hypotenuse · cost models · mean and median · tangent lines**Name / Date** \_\_\_\_\_**1 · SKILLS & IDEAS**

- S1.** Turn a margin of error into an interval; pair the opposite ends of two intervals to extremise a difference of counts.
- S2.** Chain percent changes with multipliers; compare two quantities through a common reference value.
- S3.** Express a long chain of “ $p\%$  of” statements as one multiplier on the first quantity.
- S4.** Place a figure in coordinates and use slopes or vectors for an angle between segments.
- S5.** Count intersections of  $y = a^x$  with a line by studying the minimum of the difference; boundary values matter.
- S6.** Factor a difference of cubes inside a rational expression, keeping the correct signs and domain.
- S7.** Convert cubic units with the cube of the linear factor; average parallel depths for a trapezoid.
- S8.** Use the Factor Theorem to tie a coefficient to a root; restrict by divisibility.
- S9.** Unroll a recurrence “multiply, then add” into a geometric sum; count terms carefully.
- S10.** Let one scale variable carry two ratios and solve for the total.
- S11.** Reason about monotonicity, bounds and attainable values of  $\frac{A}{1 + Bc^x}$  on a restricted domain.
- S12.** Altitude to the hypotenuse:  $\text{leg}^2 = \text{hypotenuse} \cdot \text{adjacent segment}$ ; area gives the legs.
- S13.** Build a cost function from fractions, coverage, coats and an allowance applied to one product only.
- S14.** Fix the total from the mean and the median position; push every free value to its extreme.
- S15.** Tangent line  $\perp$  radius: negative reciprocal slope, centre  $\pm$  a scaled direction vector.

**2 · REQUIRED FACTS**

- A margin of error of  $m$  points means the true percent lies in  $[\hat{p} - m, \hat{p} + m]$ ; convert to counts with that survey’s own population.
- $p\%$  greater means multiply by  $1 + \frac{p}{100}$ ;  $p\%$  less means multiply by  $1 - \frac{p}{100}$ .
- Different reference quantities: compare by dividing final values, not by subtracting percents.
- 1 yard = 3 feet, so 1 cubic yard = 27 cubic feet (the factor is cubed).
- If  $(x - r)$  is a factor of  $p(x)$ , then  $p(r) = 0$ .
- The sum of the first  $n$  terms of  $a, ar, ar^2, \dots$  is  $a \frac{1 - r^n}{1 - r}$ .
- Mean  $\times$  count = total; the median of an odd-size ordered list is its middle term.
- A tangent to a circle is perpendicular to the radius at the point of tangency.

**3 · REQUIRED FORMULAS**

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2), \quad a^3 + b^3 = (a + b)(a^2 - ab + b^2), \quad a \frac{1 - r^n}{1 - r}$$

$$\tan \theta = \frac{|\vec{u} \times \vec{v}|}{\vec{u} \cdot \vec{v}}, \quad m_{\perp} = -\frac{1}{m}, \quad AC^2 = AD \cdot AB, \quad V_{\text{prism}} = \frac{h_1 + h_2}{2} \cdot w \cdot \ell$$

## 4 · MINI THEORY SHEET

**Percent chains.** Convert every percent to a multiplier first ( $+15\% \rightarrow 1.15$ ,  $-8\% \rightarrow 0.92$ ). Choose a convenient starting value (1 or 100), push it through the chain, then compare final values by division.

**Interval extremes.** To maximise a difference  $A - B$  of two uncertain quantities, take the top of  $A$ 's interval and the bottom of  $B$ 's interval, then convert each to a count with its own base.

**Equation-counting with exponentials.** Move everything to one side, study the minimum of  $a^x - (\text{line})$ , and remember that two solutions need the constant strictly above the minimum.

**Parameters and roots.** A linear factor fixes a root; substitute it, solve for the coefficient, then run through the divisors that keep the parameter an integer.

**Optimisation with a median.** Mean gives the total. The median position is pinned; the values at or below it, and those between it and the target group, take their smallest allowed value.

**Speed shortcuts.**

| Situation                                                                 | Shortcut                                                                                                           |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| two surveys, each with its own margin<br>$p\%$ more, then $q\%$ less, ... | opposite ends of the intervals, then counts<br>multiply the multipliers; compare by division                       |
| long chain of " $p\%$ of"                                                 | set the first quantity to 1, multiply down the chain                                                               |
| angle between two segments in a figure                                    | coordinates + cross product over dot product                                                                       |
| $a^x = mx + k$ , number of solutions                                      | minimum of $a^x - mx$ ; two solutions only above it                                                                |
| $\frac{u^3 - v^3}{u - v}$                                                 | $u^2 + uv + v^2$ (no minus sign)                                                                                   |
| cubic feet $\rightarrow$ cubic yards                                      | divide by 27, not 3                                                                                                |
| $(x - n)$ a factor, $n$ and $k$ integers                                  | $p(n) = 0$ , solve for $k$ , test divisors of the constant                                                         |
| $a_{n+1} = ra_n + d$                                                      | $a_n = d \frac{1 - r^n}{1 - r}$ when $a_1 = d$ ; count doses, not days                                             |
| two ratios sharing a part<br>$\frac{A}{1 + Bc^x}$ with $0 < c < 1$        | one multiplier; one fraction-of-total equation<br>increasing, bounded by $A$ ; solve for $x$ to test attainability |
| altitude to hypotenuse                                                    | $\text{leg}^2 = \text{hypotenuse} \times \text{nearby segment}$                                                    |
| mean and median given, maximise a group                                   | total - (minimum for everything else)                                                                              |
| tangent line of slope $m$                                                 | radius direction has slope $-\frac{1}{m}$ ; centre $\pm r \cdot \text{unit vector}$                                |

## 5 · GUIDED EXAMPLES

## Example 1 — A chain of percents is one multiplier

$y$  is 20% of  $x$ ,  $z$  is 0.5% of  $y$ , and  $w$  is 300% of  $z$ . If  $y + z + w = 1.62$ , find  $x$ .

Put  $x = 1$ :  $y = 0.2$ ,  $z = 0.001$ ,  $w = 0.003$ , so  $y + z + w = 0.204$  for each unit of  $x$ .

$$x = \frac{1.62}{0.204} = \frac{135}{17}.$$

## Example 2 — Factor Theorem with an integer parameter

$p(x) = x^3 - 2x^2 + kx - 8$ , where  $k$  is a positive integer. If  $x - n$  is a factor for some positive integer  $n$ , find the sum of all possible  $k$ .

$$p(n) = 0 \Rightarrow k = \frac{8}{n} - n^2 + 2n, \text{ so } n \text{ must divide } 8: n = 1 \rightarrow k = 9; n = 2 \rightarrow k = 4; n = 4 \rightarrow k = -6; n = 8 \rightarrow k = -47.$$

Positive values: 9 and 4, so the sum is 13.

**Example 3 — Mean fixes the total, median pins one value**

Seven positive integers have mean 10 and median 8. What is the greatest possible value of the largest integer?

Total = 70. Ordered:  $x_4 = 8$ , and  $x_5, x_6 \geq 8$  (smallest choice 8);  $x_1, x_2, x_3 \geq 1$  (smallest choice 1).

$$x_7 = 70 - 3(1) - 8 - 8 - 8 = 43.$$

## 6 · PRACTICE SET — 16 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

SAT CHALLENGE · CALCULATOR

County P has 2,500 adult residents and county Q has 900. A random sample of county P residents found that 52% support a proposed bike lane, with a margin of error of 3 percentage points. A separate random sample of county Q residents found that 38% support it, with a margin of error of 4 percentage points. Based on the margins of error, what is the greatest possible value of (number of adult supporters in county P) – (number of adult supporters in county Q)?

(A) 1,033

(B) 1,069

(C) 525

(D) 958

2.

SAT CHALLENGE · NO CALCULATOR NEEDED

In 2020, the population of city M was 20% greater than the population of city N. From 2020 to 2024, the population of city M increased by 15% and the population of city N decreased by 8%. The population of city M in 2024 was  $p\%$  greater than the population of city N in 2024. What is the value of  $p$ ?

3.

SAT CHALLENGE · CALCULATOR

The positive quantities  $a$ ,  $b$ ,  $c$  and  $d$  satisfy the following:  $b$  is 0.6% of  $a$ ,  $c$  is 5% of  $b$ , and  $d$  is 150% of  $c$ . If  $b + c + d = 27$ , what is the value of  $a$ ?

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4.

SAT CHALLENGE · NO CALCULATOR NEEDED

In equilateral triangle  $ABC$ , point  $D$  lies on side  $\overline{BC}$  such that  $BD = \frac{1}{4}BC$ . What is the value of  $\tan \angle DAC$ ?

(A)  $\frac{5\sqrt{3}}{9}$   
(C)  $\frac{3\sqrt{3}}{5}$

(B)  $\frac{3\sqrt{3}}{7}$   
(D)  $\frac{\sqrt{3}}{7}$

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5.

SAT CHALLENGE · DESMOS

$$2^x = 3x + k$$

In the equation above,  $k$  is a constant. What is the smallest integer value of  $k$  for which the equation has two distinct real solutions?

(A)  $-2$ (B)  $-1$ (C)  $0$ (D)  $-3$ 

6.

SAT CHALLENGE · NO CALCULATOR NEEDED

For  $y \neq 0$  and  $y \neq \frac{1}{3}$ , which of the following is equivalent to

$$\frac{\frac{1}{y^3} - 27}{\frac{1}{y} - 3} ?$$

(A)  $\frac{1}{y^2} - \frac{3}{y} + 9$ (B)  $\frac{1}{y^2} + 9$ (C)  $\frac{1}{y^2} + \frac{6}{y} + 9$ (D)  $\frac{1}{y^2} + \frac{3}{y} + 9$

7.

SAT CHALLENGE · CALCULATOR

A raised garden bed is filled with soil. A vertical cross-section of the soil, perpendicular to the 12-foot length of the bed, is a trapezoid: the bed is 8 feet wide, and the soil is 1.5 feet deep along one side and 3.5 feet deep along the opposite side, with the surface sloping uniformly between them. Soil is sold at \$45 per cubic yard and billed for the exact volume (1 yard = 3 feet). What is the cost, in dollars, of the soil needed?

8.

SAT CHALLENGE · NO CALCULATOR NEEDED

The polynomial  $p(x) = x^3 - 9x^2 + kx - 12$  has a positive integer constant  $k$ . For some positive integer  $n$ ,  $x - n$  is a factor of  $p(x)$ . What is the sum of all possible values of  $k$ ?

9.

SAT CHALLENGE · CALCULATOR

A patient takes a 100-milligram dose of a medication at 8:00 a.m. and again at 8:00 p.m. every day, beginning at 8:00 a.m. on Monday. The dose is absorbed instantly, and between any two consecutive doses exactly half of the medication present in the body is eliminated. To the nearest tenth of a milligram, how much medication is in the body immediately after the 8:00 a.m. dose on Thursday?

(A) 187.5

(B) 196.9

(C) 198.4

(D) 199.2

10.

SAT CHALLENGE · NO CALCULATOR NEEDED

A box contains only red, green and blue flash drives, 140 in all. The ratio of red drives to green drives is 7 to 4. The ratio of blue drives to drives that are not blue is 3 to 11. How many more red drives than blue drives are in the box?

11.

SAT CHALLENGE · CALCULATOR

The function  $P(t) = \frac{120}{1 + 5(0.8)^t}$  models the number of active users of an app, in thousands,  $t$  weeks after launch, for  $0 \leq t \leq 10$ . Consider the following statements.

- I.  $P$  is increasing on the interval  $0 \leq t \leq 10$ .
- II. There is a time  $t$  in the interval at which  $P(t) = 60$ .
- III. There is a time  $t$  in the interval at which  $P(t) = 80$ .

Which of the statements must be true?

- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II, and III

12.

SAT CHALLENGE · NO CALCULATOR NEEDED

In right triangle  $ABC$ , the right angle is at  $C$  and  $\tan A = \frac{5}{12}$ . The area of the triangle is 120. Point  $D$  lies on  $\overline{AB}$  such that  $\overline{CD} \perp \overline{AB}$ . What is the length of  $\overline{AD}$ ?

- (A)  $\frac{50}{13}$
- (B)  $\frac{288}{13}$
- (C)  $\frac{120}{13}$
- (D) 24

13.

SAT CHALLENGE · CALCULATOR

The function  $f$  is defined by  $f(x) = a \cdot b^x$ , where  $a$  and  $b$  are positive constants. For every value of  $x$ ,  $f(x + 2)$  is 44% greater than  $f(x)$ . For every value of  $x$ ,  $f(x + 3)$  is  $p\%$  greater than  $f(x)$ . What is the value of  $p$ ?

(A) 132

(B) 60

(C) 66

(D) 72.8

14.

SAT CHALLENGE · CALCULATOR

A contractor paints the walls of a room with total wall area  $x$  square feet. Primer is applied to all of the wall area in one coat; one gallon of primer covers 400 square feet and costs \$20. Then paint is applied to  $\frac{3}{5}$  of the wall area in two coats; one gallon of paint covers 300 square feet per coat and costs \$30. The contractor buys 10% more paint than the calculated amount, but buys exactly the calculated amount of primer. Gallons may be purchased in fractions. Which of the following functions gives the total cost  $C(x)$ , in dollars, of the primer and paint?

(A)  $C(x) = 0.182x$ (B)  $C(x) = 0.187x$ (C)  $C(x) = 0.232x$ (D)  $C(x) = 0.170x$

15.

SAT CHALLENGE · NO CALCULATOR NEEDED

A list of nine positive integers has a mean of 12 and a median of 9. What is the greatest possible value of the sum of the three largest integers in the list?

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16.

SAT CHALLENGE · NO CALCULATOR NEEDED

$$x^2 + y^2 - 4x + 6y - 12 = 0$$

In the  $xy$ -plane, the graph of the equation above is a circle. Line  $\ell$  has slope  $\frac{3}{4}$  and is tangent to the circle at a point with a positive  $y$ -coordinate. What is the  $y$ -intercept of  $\ell$ ?

(A) 1

(B)  $\frac{7}{4}$ (C)  $-\frac{1}{3}$ (D)  $-\frac{43}{4}$ 

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